

Folding from predicted contacts instead of MSAs

Helico + Marifold · results on 91 paired FoldBench monomers

github.com/Open-Athena/helico · PR #13, issue #11

The idea

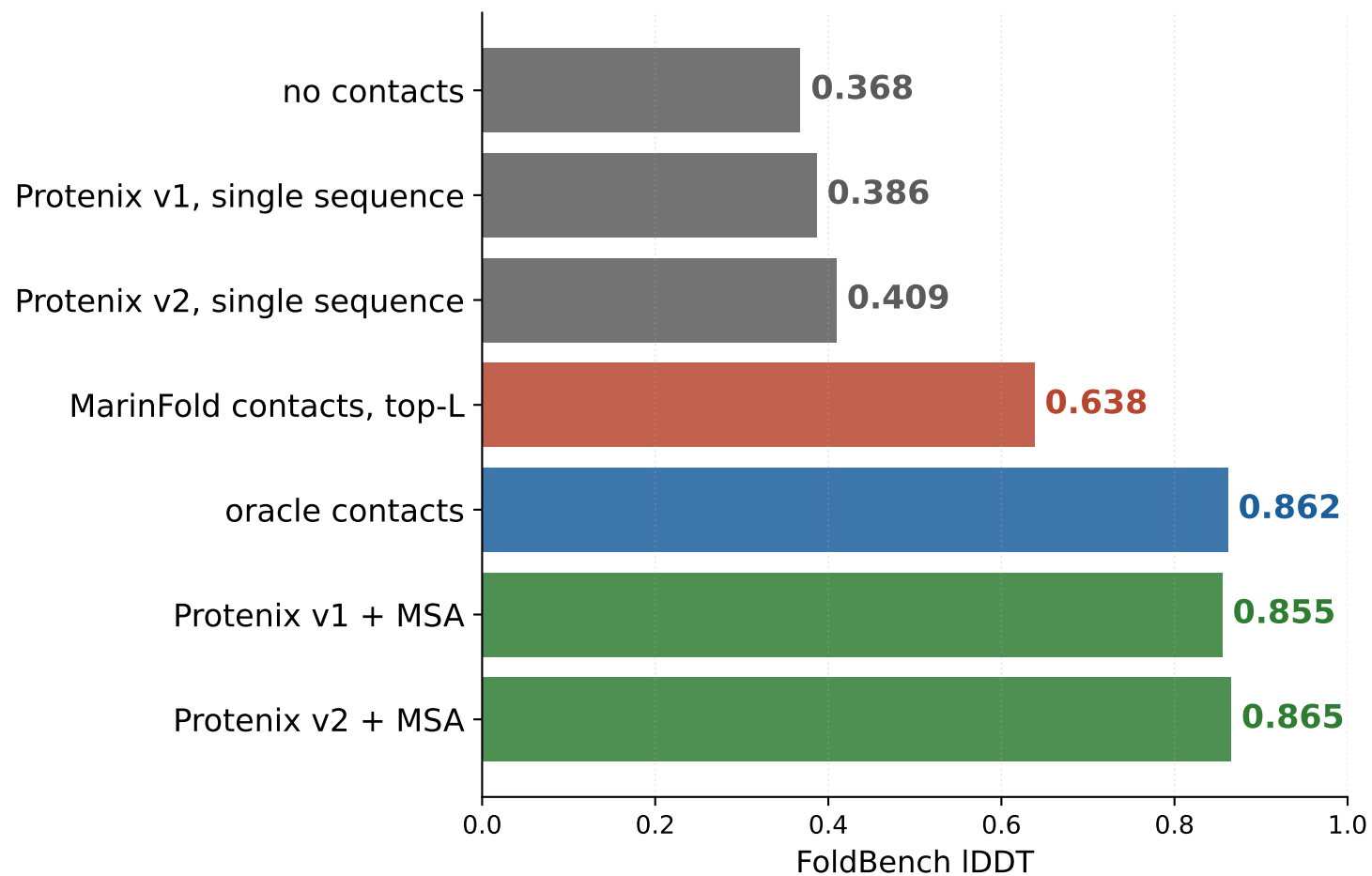
AF3-family models lean on MSAs. Strip the alignment and accuracy collapses.

- MarFold predicts residue-residue side-chain contacts directly.
- Feed those contacts to the folding model in place of the alignment, and the MSA search leaves the critical path.
- Helico takes a three-state contact matrix — contact / no-contact / unknown — added into the pair representation.
- Training samples the conditioning level per example, so one model spans no contacts through fully specified.

*Helico arms use no alignment and no conservation profile, at training or inference.
The Protenix +MSA baselines do use MSAs -- that is the comparison.*

Where things land

FoldBench monomer subset, 91 paired targets · Helico arms use no MSA; Protenix +MSA arms do

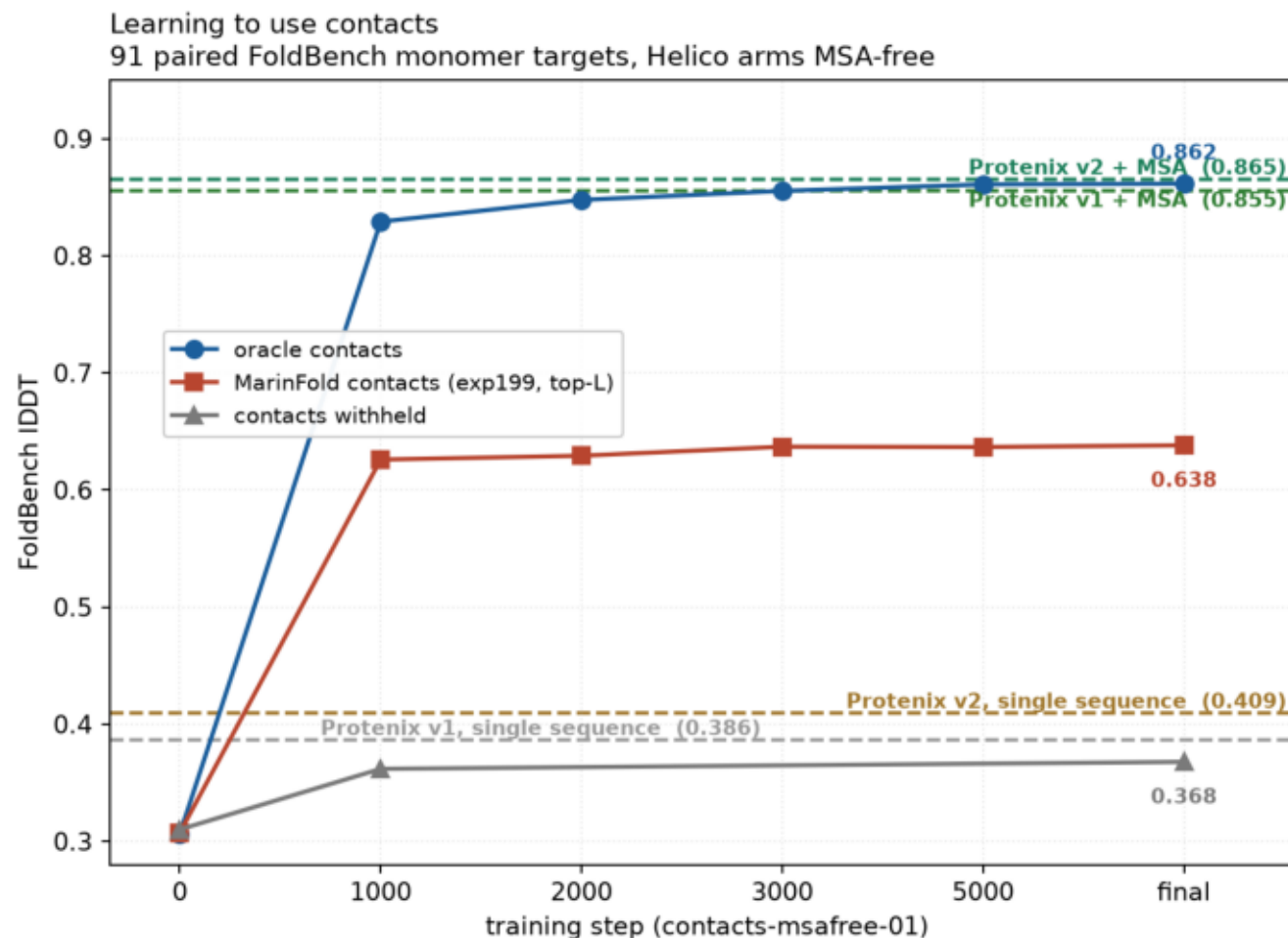


**Contacts beat
single sequence
by a wide margin**

**...but do not reach
MSA-level accuracy**

How the model learned to use contacts

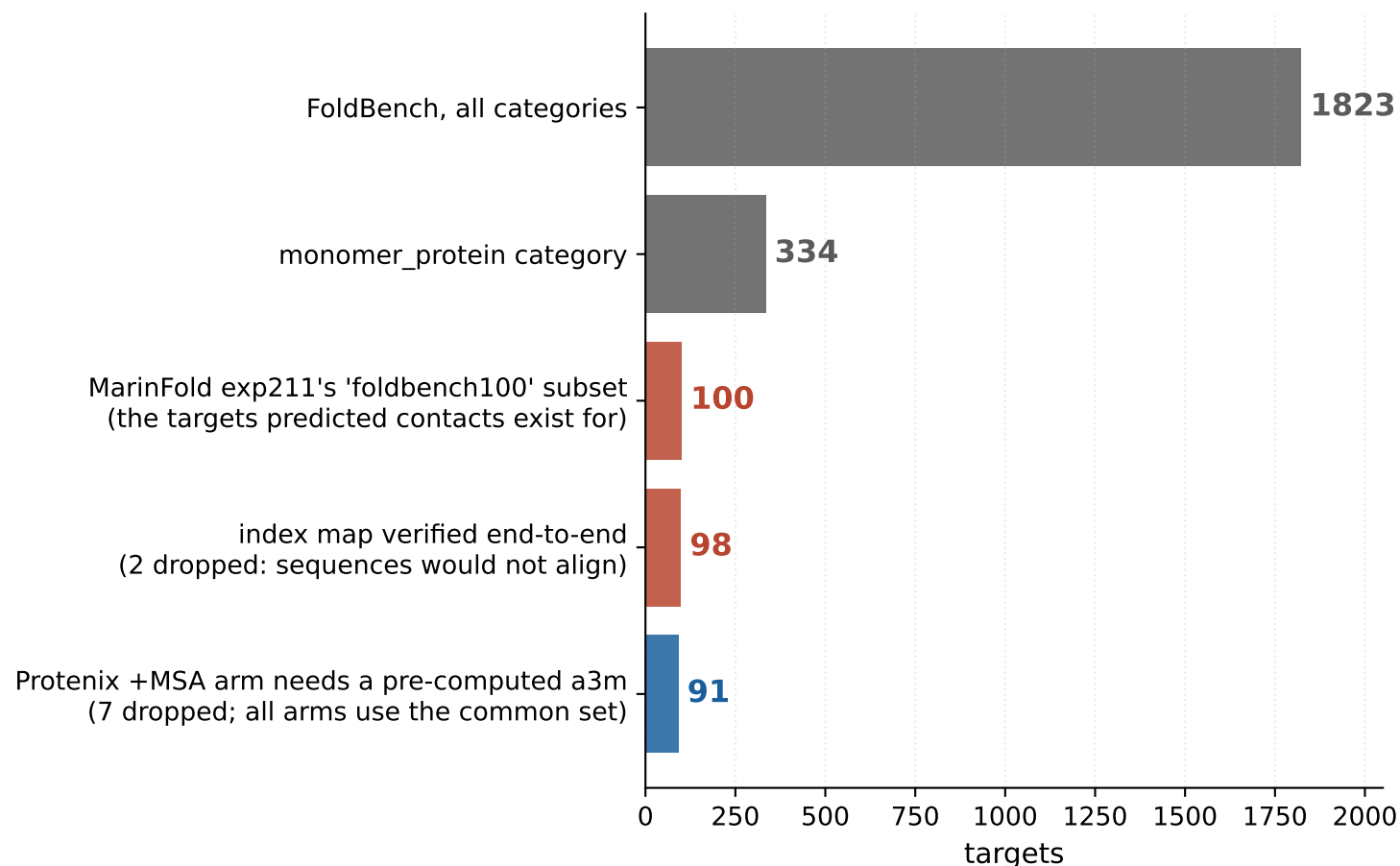
checkpoint sweep of contacts-msafree-01, benched on MarinFold contacts at each step



Step 0 is the warm start -- Protenix v1 weights, contact projection still zero-initialised, so conditioning is an exact no-op. All three arms coincide there within 0.003. Almost all of the learning happens in the first 1000 steps.

What we benchmark on, and how it was chosen

FoldBench is far bigger than this subset -- here is the whole chain



Monomers only -- so these numbers are not comparable to the assembly-set results reported earlier.

The full target list is in [experiments/marinfold_contacts/arms/targets.csv](#).

Yes: predicted contacts clearly beat single-sequence folding

vs Protenix v2 -- the stronger baseline -- in single-sequence mode

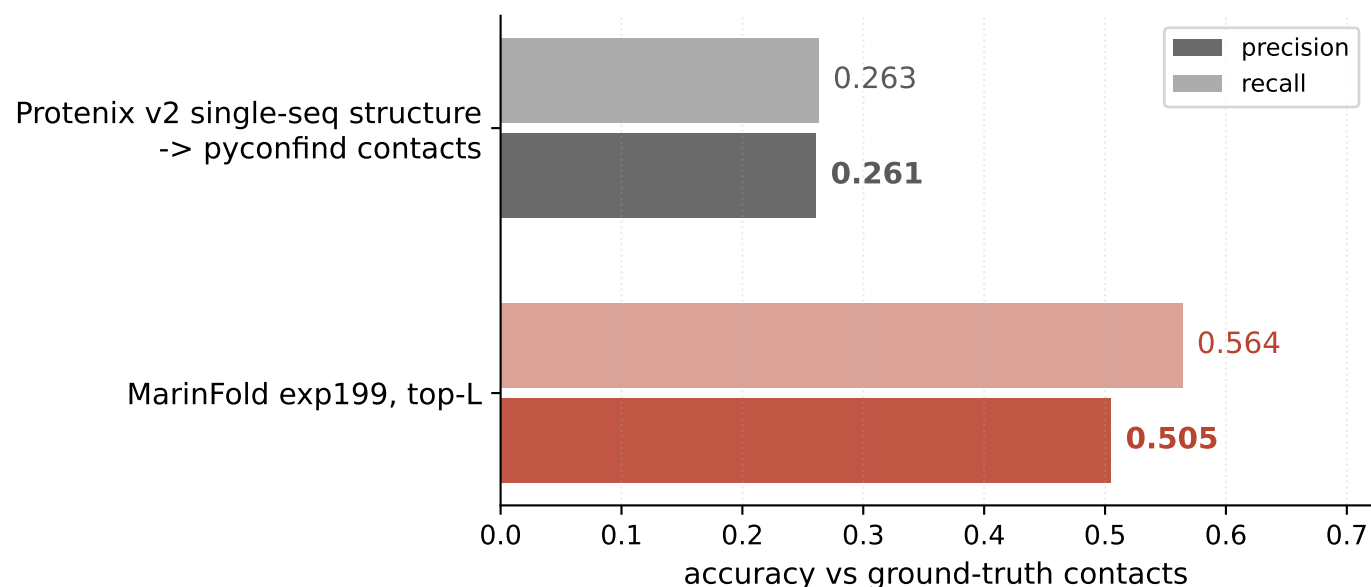
- **MarinFold contacts (top-L): $+0.229 \pm 0.022$ IDDT (t=10.4, better on 76/91)**
- Protenix v2 is genuinely the harder baseline: +0.023 over v1 without MSAs, +0.010 with them. The margin survives anyway.
- Holds at every contact budget, and even a single rollout beats it.

The control that makes this credible

Our own contacts-off arm scores 0.368 — BELOW Protenix v2's 0.409 (-0.042 ± 0.008). The fine-tuned model has no intrinsic edge with no information. All of the gain comes from the contacts.

Is MarinFold actually supplying better contacts?

vs reading contacts straight off Protenix v2's own single-sequence structure



Both emit a comparable number of contacts -- 265 vs 270 per target, against 261 true ones -- so this is a like-for-like comparison. MarinFold is ~1.9x more precise and ~2.1x higher recall.

So the gain over single-sequence folding is not Helico extracting more from equivalent information: the contact map really is better.

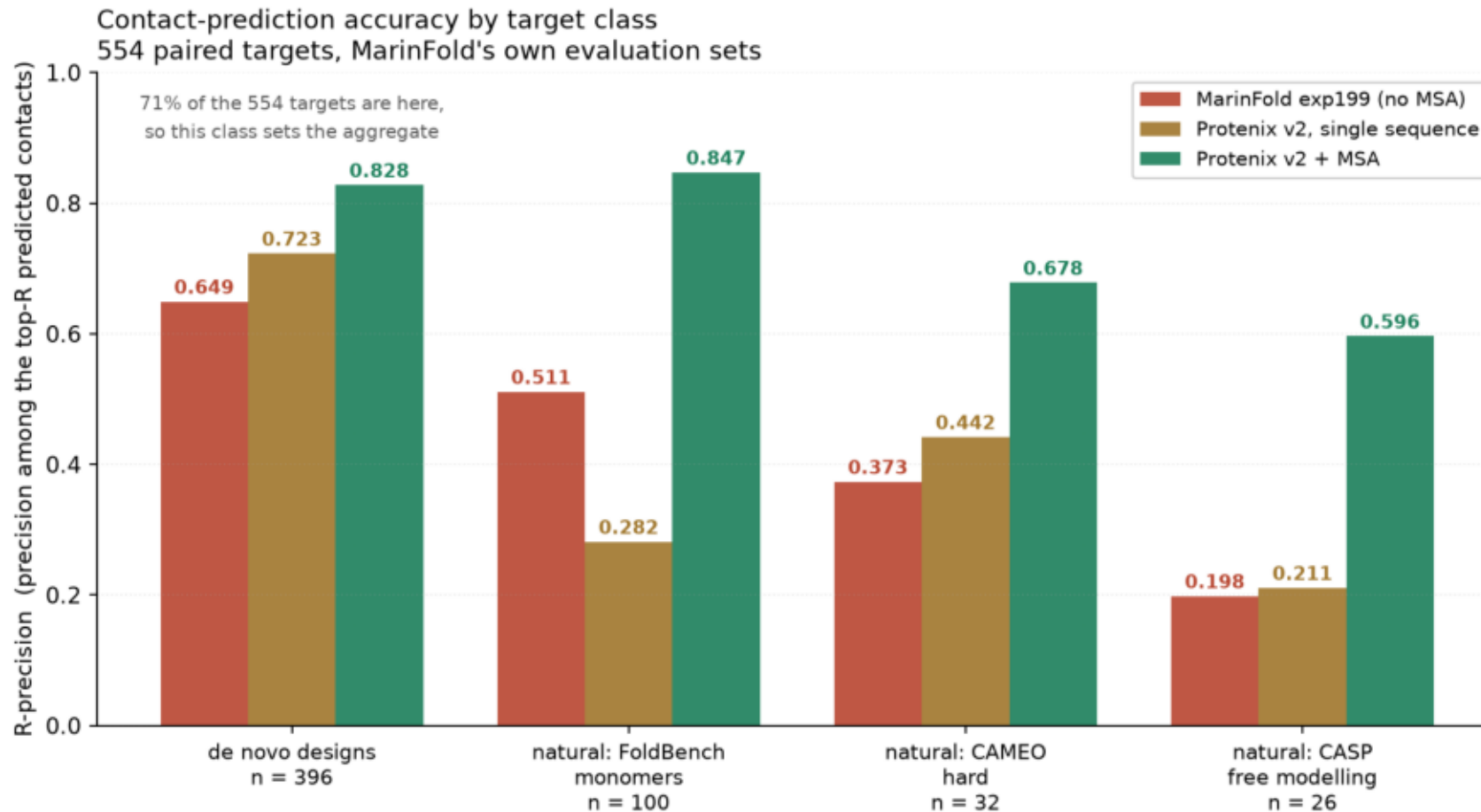
The remaining gap to MSAs is contact quality

MarinFold top-L 0.638 -> Protenix v2 + MSA 0.865

- Oracle contacts reach 0.862 against Protenix v2 + MSA's 0.865. That difference, +0.004, is not statistically distinguishable from zero.
- A perfect contact map is therefore worth as much as an alignment to the best model available. Nothing about the approach caps out below MSAs.
- The entire shortfall is the quality of the contacts we can currently predict -- 0.638 with predicted ones versus 0.862 with perfect ones.

But MarinFold's own numbers say it ties Protenix v2 SS

R-precision by target class -- both statements are true



Pooled over all 554: MarinFold 0.587 vs Protenix v2 SS 0.603 -- the tie. 71% of those targets are designed proteins, where Protenix v2 SS is much the better contact predictor. MarinFold's advantage is confined to foldbench100 — it also loses on CAMEO hard and ties on CASP FM, both natural. The folding result inherits that limit.

Was MarInFold trained on these proteins?

mmseqs search of the distillation corpora against the 98 targets

- AFDB distillation corpus: 76/98 targets have a homolog at $\geq 25\%$ identity, 27 at $\geq 50\%$. So yes, homologs are present.
- But identity does not predict accuracy: $r = -0.044$ between a target's best homolog identity and its IDDT.
- Dropping every target with a $\geq 50\%$ homolog STRENGTHENS the result: +0.259 versus +0.229 on the full set.

Still open

The ESM-Atlas scan is still running. An earlier coarse check reported 2/98 and was wrong — it is superseded by the mmseqs numbers above.

What to do about it

- Retrain with false positives sampled from near-miss pairs rather than uniformly. The model has never seen the error distribution it faces. Cheapest lever, and it is on our side rather than MarFold's.
- Vote aggregation across rollouts is worth $+0.071 \pm 0.008$ over a single rollout at the same budget — keep using it, never mix the recipes.
- Improve MarFold's error profile, not only its precision/recall.

Caveats

Warm-started from Protenix v1. Monomers only, so not comparable to the earlier assembly-set numbers. MarFold indexes into the published prompt, Helico into resolved residues — only 15/100 agree outright; the map is verified by round trip at Jaccard 0.998.